

# **Student Development Platform**

## **Requirements Modeling**

## Table of Contents

|                                                                 |    |
|-----------------------------------------------------------------|----|
| Introduction .....                                              | 3  |
| System Requirements .....                                       | 5  |
| 1. Student Profile Management .....                             | 5  |
| 1.1 User Registration and Authentication .....                  | 5  |
| 1.2 Profile Updates .....                                       | 5  |
| 2. Personalized Learning Path Creation .....                    | 5  |
| 3. Counseling and Mental Health Support.....                    | 6  |
| 4. Social and Emotional Learning (SEL) Program Management ..... | 6  |
| 4.2 Evaluation and Reporting.....                               | 6  |
| 5. Project and Task Management.....                             | 7  |
| 5.2 Task Management .....                                       | 7  |
| 6. Communication and Notifications.....                         | 7  |
| 6.1 Messaging System.....                                       | 7  |
| 6.2 Notification System.....                                    | 7  |
| The structural Model.....                                       | 9  |
| Functional Model .....                                          | 11 |
| Behavioral Model .....                                          | 13 |
| UML DIAGRAMS .....                                              | 15 |
| Use case diagram.....                                           | 15 |
| Use case description .....                                      | 16 |
| Sequence diagrams .....                                         | 22 |
| Class Diagram .....                                             | 24 |
| References.....                                                 | 25 |

## **Introduction**

With the current changes and progressing advancement of educational systems, facilitating the indoctrination of comprehensive student development is crucial. The AI-Driven Holistic Student Development System comes out as a revolutionary innovation aimed at the achievement of the educational objectives through employing AI and machine learning technologies to develop wellrounded persons. It is a new system that targets and concentrates on the academic achievement, learning behavior, and the fulfillment of a learner.

In turn, the data gathered from various sources are conglomerated in the system to form a holistic student profile. Programs of AI algorithms then use this developed profile to recommend content and learning materials, create learning tracks, and propose assistance. In addition, the system ensures that mental care can be directly accessed at any time with the help of recommending early intervention measures besides having around the clock counseling through artificial intelligence. Thus, active participation helps students develop social and emotional learning (SEL) through encouraging communication, collaboration, and emotions' understanding as well.

This system also does not limit itself to the classroom only as we will be seeing shortly. Using AI-based algorithms, students are encouraged to engage in complementary activities leading to personal development and enhancement of their college experience. Furthermore, the system offers AI-supported lesson planning tools for lecturers as well as holistic counselors' dashboards with the necessary tools to ensure correct and efficient working.

Therefore, the AI-Driven Holistic Student Development System enhances academic performance, personal and social-emotional health and well-being together with a comprehensive human-beings development, preparing students to become contestants in the 21st century.

## **Background**

The purpose of this student development system is mostly so that the student may develop in all the civil aspects as may be deemed academically, health-wise, co-curricular activities, or almost all aspects they may consider requisite in the student's growth. This is why AI and machine learning have been applied in this system; for example, assisting in offering individual assistance when trying to understand certain interested areas concerning every learner. The incorporation of AI and machine learning in the system has the following benefits help the learners in the following manner. The development of this system academically will help students choose subjects that they like while it offers more ways to learn the subject matter more deeply. Mental Health and Well-being also have to come first which is achieved through access to AI-driven counseling 24/7 and an early warning system that would quickly pick up any issues hence ensuring prompt action is taken or given if required. By so doing, Social and emotional learning is highly enhanced by the acquisition of vital skills e.g. empathy, emotional intelligence as well as effective communication thereby fostering improved socialization as well as having a supportive school climate.

Through the system it provides comprehensive student profiles that consolidate data from academic, extracurricular, and social activities, ensuring balanced growth across all areas. Through the system the learning resources are made more efficient, to the lectures and the administrators making informed decision on optimized learning resource. Finally, it makes students ready for long and continued learning process in other educational systems to overcome life challenges that they are bound to encounter in their respective educational systems in their lifetime.

# **System Requirements**

## **1. Student Profile Management**

### **1.1 User Registration and Authentication**

- The system shall provide means for creation of such profiles as student's personality profile containing personal information about the student, his or her academic background, including school records, participation in extracurricular activities, disciplinary and academic records, and medical records.
- The system shall be capable of logging students, lecturers, counselors, and administrators in by ID with a Password.
- It shall then check their credentials and, if valid, allow user access to the appropriate dashboards.

### **1.2 Profile Updates**

- The system shall have an option through which counselors can add, remove or edit health information that related to the learners and record counseling sessions.

## **2. Personalized Learning Path Creation**

**2.1 Learning Preferences and Interests** • It will enable the students to have the ability to set and modify their preferences of the course and interests of their learning.

- The system shall employ the use of AI to predict the course of previous academic outcomes and preferences in order to recommend appropriate learning modules that should be taken by the learner at the next level.

**2.2 Resource Management** • A feature of the system shall allow the lecturers to upload content like books, articles, videos, and tests as well as updates them.

- The system shall recommend to the lecturers other AI suggested learning resources which the lecturers may further add in their modules.

**2.3 Learning Path Customization** • The system shall generate personalized learning paths for students based on their profiles and preferences.

- The system shall allow learning paths to be altered based on the new data that may be collected or as the preferences change.

### **3. Counseling and Mental Health Support**

**3.1 Session Management •** The system shall enable counsellors to book appointments for special sessions with students.

- The system shall also have provisions to record session details with the details that may include date, time, and any other comment.

**3.2 Mental Health Monitoring •** The system shall incorporate Artificial Intelligence that shall be used to identify the signs of stressed, anxious and other related mental health conditions from body and gestural language of the students.

- The system shall alert counselors if any mental health indicators are triggered.

**3.3 Support Resources •** It will offer details on who and where students can turn to seek mental health support services and other resources, the system shall.

- The system should be equipped with an instant messaging window for the students where they can get prompt help from the system; the system should also be capable of forwarding the students to a counselor if required.

### **4. Social and Emotional Learning (SEL) Program Management**

**4.1 Program Creation and Management •** The system shall allow SEL program managers to create and update interactive SEL programs and activities.

- The system shall log student participation in SEL activities and track progress.

### **4.2 Evaluation and Reporting**

- Thus, the system shall incorporate; The system will conduct periodic tests to measure the students' emotional intelligence, and their social competencies.
- The system shall generate progress reports and integrate them into student profiles.

## **5. Project and Task Management**

**5.1 Project Creation and Assignment •** The system shall allow project leaders and club representatives to create and manage projects, including setting timelines and assigning teams.

- The system should allow students to self-organize their learning and build projects with support from classmates.

### **5.2 Task Management**

- The assigned system shall provide project leaders with the options of creating tasks in a project, defining the respective difficulty level for these tasks, as well as defining the status of their completion.
- This web-based system should be able to manage the status, update and alert the team members concerning the due dates.

## **6. Communication and Notifications**

### **6.1 Messaging System**

- The system shall have a chat interface enabling students, counselors, lecturers, and administrators to communicate.
- The system shall also store all messages in a log which can be referred back at a later date.

### **6.2 Notification System**

- It shall enable the system, the student, the instructor, and/or the counselor to send notifications as regards important updates such as the time left before a particular deadline, and/or a reminder for a counseling session, or other important notices in the system.

### **Nonfunctional requirements**

- The system shall be able to response without delay, and every action shall be performed in less than 2 seconds at maximum loading.
- It should sufficiently accommodate over 1000 users to be able to use the system at the same time without compromising the organizations performance.
- The system shall be designed to have the ability to be horizontally as well as vertically scalable dependent on the growth in user numbers or data sizes.
- The system shall also meet an up time of 99.9%; it means that the system shall be available for the users most of the times.
- There shall be redundancy in access of the system to prevent interruption in the event that a server crashes.
- The system shall employ data encryption at the transmission and rest phase to enhance security of the data.
- This shall be through maintaining high-security measures on the system by using role-based access control mechanisms.
- All dealings related to the system shall be recorded as to help auditors with records of users' activities.
- It shall also have the ability and flexibility to accommodate multiple interfaces that can be used by the users..
- The system shall respect privacy of data and the user must avail his consent to data processing activities.

## **The structural Model**

The Student Development System was created to comprehensively assist students of private universities. This data structure is highly planned and divided into several primary categories that support business operations in the organization such as student profiles, learning resources, counseling and mental health data, SEL programs, system users, project and task management, and communication and notifications. The student profiles include personal information (student ID, name, age, grade level, contact information), their academic records (grades achieved in different subjects taken within a particular period of time and standardized test scores), extracurricular activities and also data that they participate in behavioral data (records of social interactions and disciplinary actions) and health records (including notes from counselors and medical professionals).

Learning resources include course content, which is in the form of notes, worksheets, and assignments, among others, under the charge of lecturers. Counseling and mental health data include reports of counseling and associated mental health markers and identified resources. SEL programs indicate the specific events that involve the practice of interpersonal skills and fun tasks and exercises that increase empathy. Users in the system include students, instructors, counselors and other staff, with their profiles containing info in the roles and span of access. Project management data of project and task control contains basic information such as project description, timeframes, identified teams and completed work.status. Due to the high relative importance of communication and notifications, it captures messages as well as alerts for the deadlines or appointments.

Objects created and used by the system include student objects (attributes: student ID, name, age, grade, academic records, extracurricular activities, behavioral data, health records; methods: update profile, add academic record, log extracurricular activity, record behavior, update health record), Lecture objects (attributes: lecture ID, name, module(s) taught, contact information; methods: assign grades, upload learning resources, update learning materials), counselor objects

(attributes: counselor ID, name, qualifications, contact information; methods: schedule session, log session notes, update mental health indicators), learning resource objects (attributes: resource ID, title, type, subject, level, URL/location; methods: add resource, update resource, delete resource), assignment objects (attributes: assignment ID, title, description, subject, due date, assigned students; methods: create assignment, update assignment, grade assignment), assessment objects (attributes: assessment ID, type, subject, date, student scores; methods: create assessment, update assessment, record scores), SEL program objects (attributes: program ID, title, description, activities included, participants; methods: create program, update program, track progress), project objects (attributes: project ID, name, description, start date, end date, assigned teams, phases; methods: create project, assign team, add phase, update status), task objects (attributes: task ID, name, description, difficulty rating, assigned teams, status; methods: assign task, update task status, mark task complete), and notification objects (attributes: notification ID, type, recipient, content, timestamp; methods: send notification, update notification, delete notification).

This comprehensive data structure ensures that all relevant information is captured, maintained, and utilized effectively to enhance the educational experience and overall well-being of students.

## **Functional Model**

The AI-Driven Holistic Student Development System operates through a series of interconnected functionalities that cater to the various actors involved. Here's a breakdown of the system's core functions and interactions:

### **Data Collection and Integration:**

**Students:** The system pulls information about the student through exam, questionnaires, and they are also encouraged to report information about themselves.

**Lecturers:** They feed in grades, performance details and any observation they may have of the learning process into the system.

**External Systems:** The soft links with other existing systems used to collect information about students make it possible to import data into the system (for instance, attendance records, demographic information).

#### Data Analysis and Insights Generation:

This study aims at identifying these alternative narratives, that is, the specific texts and discourses that could hold the potential to reframe the conversation and the shared meaningmaking process.

Such data is then processed by the employed AI algorithms in order to learn new patterns and anticipate students' requirements as well as to adapt the learning processes.

This evaluation offers information about academic achievement, possible areas of teaching/learning difficulty, mental health markers, and socio-emotional growth.

### **Personalized Learning and Support:**

**Students:** Depending on the data collected, students get suggested sources, teaching methods, and learning activities based on individual student's needs and learning goals.

**Lecturers:** The system provides teachers with intelligent lesson planning tools, which help to plan lessons with focus on differentiated students' requirements. Moreover, the teachers are also notified and get suggestions to look into particular students who need assistance.

**Counselors:** They use the student's progress panel in tracking the student's progress to highlight the deserving and needy students who should be assisted. The system can also offer appropriate resources and assist in support strategies.

### **Mental Health and Well-being Support:**

It is therefore elemental to recognize A. I. as a tool and to steer the focus towards how we can humanize it.

**Students:** The system also provides optional integration with artificial intelligence-based counseling applications for avocational self-help and mental health assessment, respectively.

**Counselors:** They can use it as a tool to track such things as students' well-being indicators and respond to them as appropriate. This system can also refer students to right mental health services as per the circumstance.

### **Other requirements:**

The Social and Emotional Learning (SEL) Program Management allows for the creation and updating of SEL programs, tracking student participation, and evaluating emotional intelligence and social skills.

Project and Task Management supports the creation and assignment of projects and tasks, allowing for collaboration and tracking of task completion. The Communication and Notifications system provides a messaging interface for all users and generates notifications for important updates.

**System Administrators:** They are responsible for user control and restrictions regarding accessing and sharing information. Furthermore, they oversee the performance of the systems, identify problems and resolve them, as well as integrating changes in the system with enhanced features.

## **Behavioral Model**

The Student Development System operates dynamically to support various business processes through a series of interactions and collaborations among system objects. This section provides an in-depth explanation of these interactions and the internal workings of the system to fulfil its functional requirements.

The student registration officer initiates the process by entering student details into the registration form. This form includes attributes such as registration ID, age, academic records, extracurricular activities, previous behavioral records, and health records. Upon submission, the registration form creates or updates a student profile in the database. Students can log in through the Admin Portal and update their learning interests and social interactions weekly. The system dynamically updates the student profile with new data, ensuring it is comprehensive and current.

After a student logs in with his user name and password, the student profile server will authenticate and let the student login. Then the adaptive learning algorithm continuously analyses data from the student profile, including academic performance, learning interests, and progress in subjects. Based on this analysis, the algorithm customizes learning paths by selecting relevant learning resources, such as books, videos, and interactive exercises.

Lecturers manage and update these resources within the system, ensuring they are tailored to each student's needs. The system delivers these customized resources to the students, adapting over time as more data is gathered.

Counsellor's log into the counselling portal to access their assigned user profiles. The session log records details of counselling sessions, including date, time, and session notes.

Mental health indicators are updated based on AI-driven analysis of student behaviour and interactions. The system alerts counsellors to any critical changes in mental health indicators, prompting timely interventions. A student can seek help through chatbot at any time that they need help. And the AI chat bot is directly notifying a counselor in charge at the time.

The system evaluates students' emotional intelligence and social skills through SELE evaluations.

Progress reports are generated and integrated into the student profile to track development over time.

For a lecture, they can upload their learning materials to the learning resource server system by logging in with their username and password. They will also get AI-modified learning options that they can include in their modules to increase the quality value of the system.

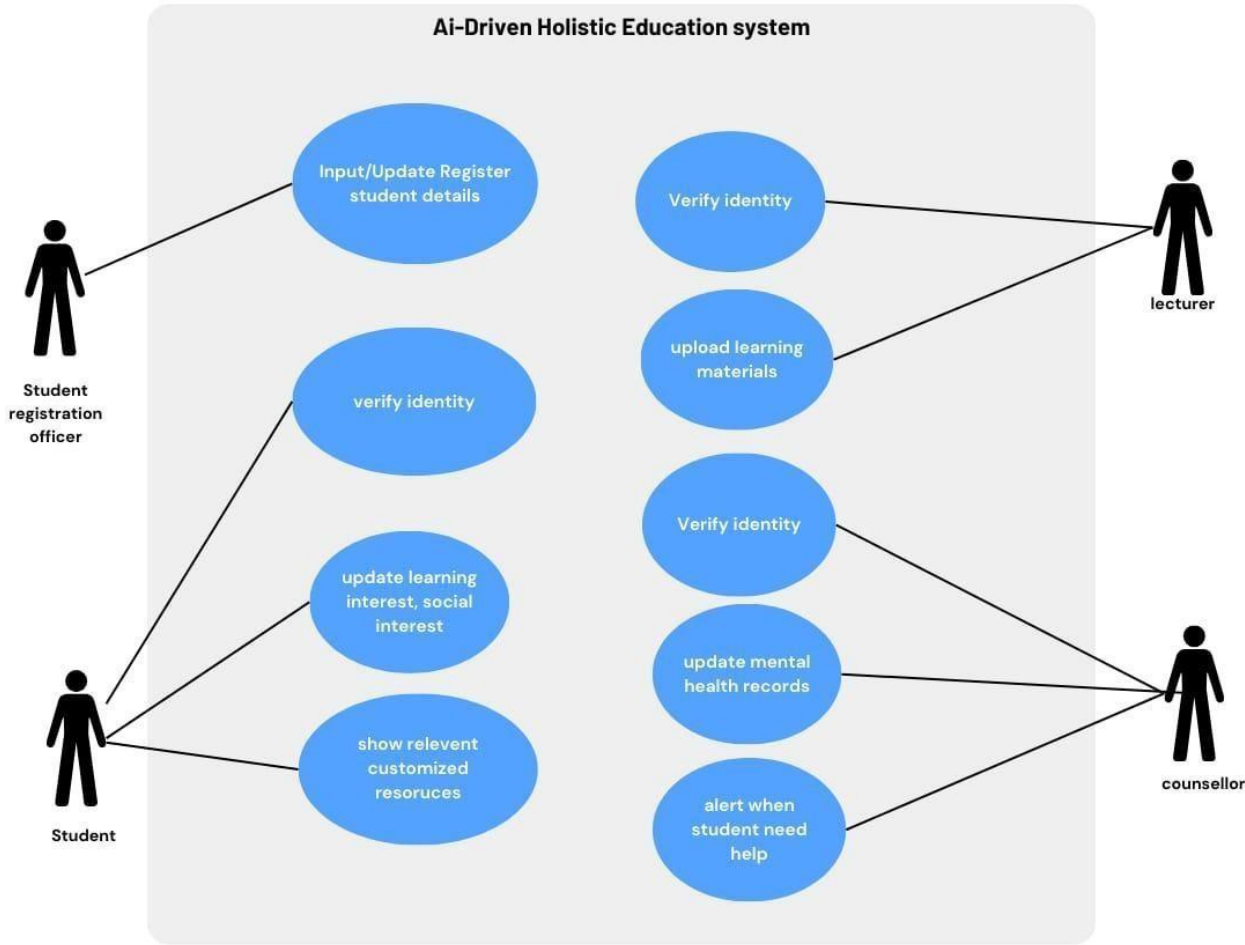
A club representative can log into the system and update their ongoing projects. Students participate in these projects according to their wish

A student can also create their own projects and add multiple students through the system in order to create and maintain a project according to their idea. The project leader creates projects and assigns tasks using the project and task objects. Each project includes a project ID, description, start and end dates, assigned teams, and phases. Tasks are detailed with IDs, descriptions, difficulty ratings, and statuses, which are updated as work progresses. Notifications are sent to relevant students and team members about deadlines and status updates.

Messages and notifications are essential for maintaining effective communication among students, counsellors, lecturers, and administrators. The system automatically generates alerts for upcoming deadlines, counselling appointments, and other critical updates. User Profiles store contact information, ensuring notifications are delivered accurately and promptly.

UML DIAGRAMS

Use case diagram



## Use case description

**Use Case Name:** Get Information on Social Interaction and Mental Health Monitoring –  
**ID:**01

**Importance Level:** High

**Stakeholders and Interests:**

- **Student:** Wants to receive personalized support and interventions for their social and mental well-being.
- **Counselor:** Needs timely and accurate information to provide effective counseling and support to students.
- **Lecturer:** Interested in understanding students' social engagement and potential mental health issues affecting their academic performance.
- **Administrators:** Need to ensure the system is effectively monitoring and supporting student well-being.
- **AI System:** Provides analysis and alerts based on student interactions and behavior data.

**Brief Description:**

This use case describes the process through which the system monitors and provides insights into students' social interactions and mental health status. It involves logging student activities, analyzing interaction data, detecting potential mental health issues, and notifying counselors for intervention.

**Trigger:**

A student's interaction with the system, such as logging in, updating their profile, or engaging in activities.

**Relationships:**

**Association:** Student, Counselor, AI System

**Include:** Log Interactions, Analyze Data, Generate Reports, Notify Counselor

**Extend:** Mental Health Monitoring, Social Interaction Analysis

**Normal Flow of Events:**

- Student logs into the system and engages in various activities.
- System records these interactions, noting participation and engagement levels.
- AI system analyzes recorded interactions to assess social engagement.
- AI system monitors behavior for signs of stress or mental health issues.
- System generates a detailed report on social interactions and mental health indicators.
- System alerts counselor if any mental health issues are detected.
- Counselor reviews report and schedules a session with the student

**Use Case Name:** Personalized Learning Path Creation

**ID:**02

**Importance Level:** High

**Primary Actor:** Student

**Use Case Type:** Main

**Stakeholders and Interests:**

- **Student:** Seeks a customized learning experience tailored to their preferences and academic needs.
- **Lecturer:** Wants to provide appropriate resources and support for each student's learning path.
- **AI System:** Responsible for analyzing data and generating personalized learning paths.
- **Administrators:** Need to ensure the system effectively supports personalized learning.–

**Brief Description:**

This use case describes the process of creating a personalized learning path for each student. It involves collecting student preferences, analyzing academic performance, and recommending suitable learning modules and resources.

**Trigger:**

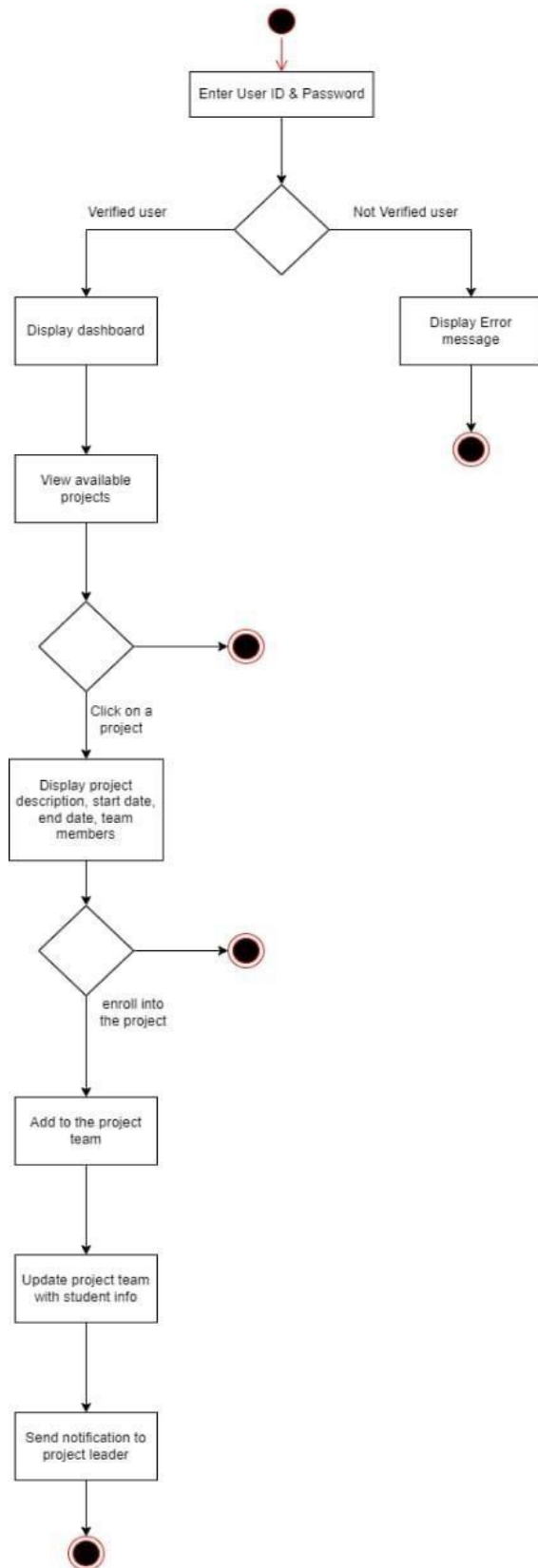
A student logs into the system and updates their learning preferences, or new academic performance data is available.

**Relationships:**

- **Association:** Student, Lecturer, AI System
- **Include:** Update Preferences, Analyze Data, Recommend Modules
- **Extend:** Resource Management, Learning Path Adjustment

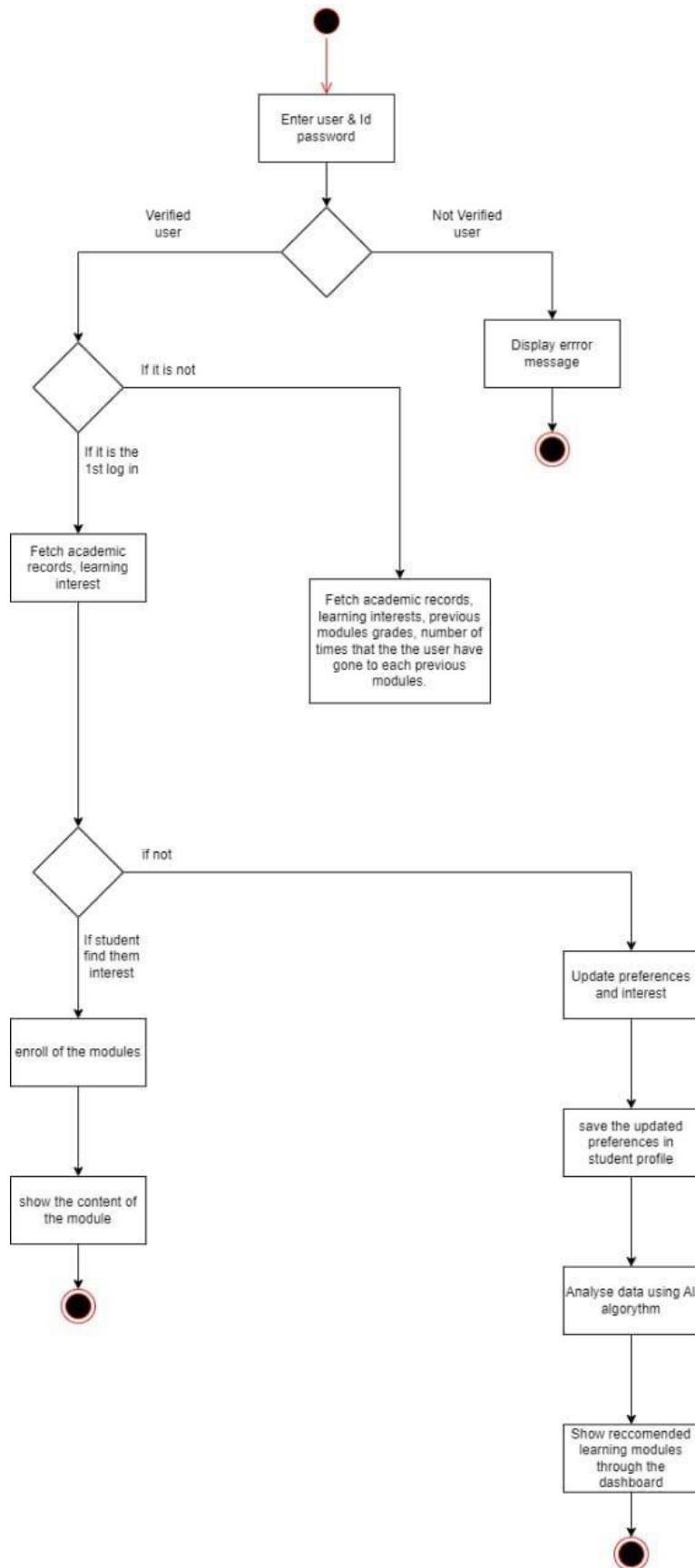
**Normal Flow of Events:**

- Student logs into the system and updates their learning preferences and interests.
- System records these updates in the student's profile.
- AI system analyzes the student's academic performance, learning preferences, and past interactions.
- AI system recommends personalized learning modules and resources such as books, articles, videos, and exercises.
- System generates a personalized learning path for the student based on the analysis.
- Lecturer uploads and updates recommended learning materials.
- System integrates lecturer's resources with AI-recommended materials.
- Personalized learning path is delivered to the student.
- Student starts following the learning path, engaging with the resources provided.



## Activity diagrams

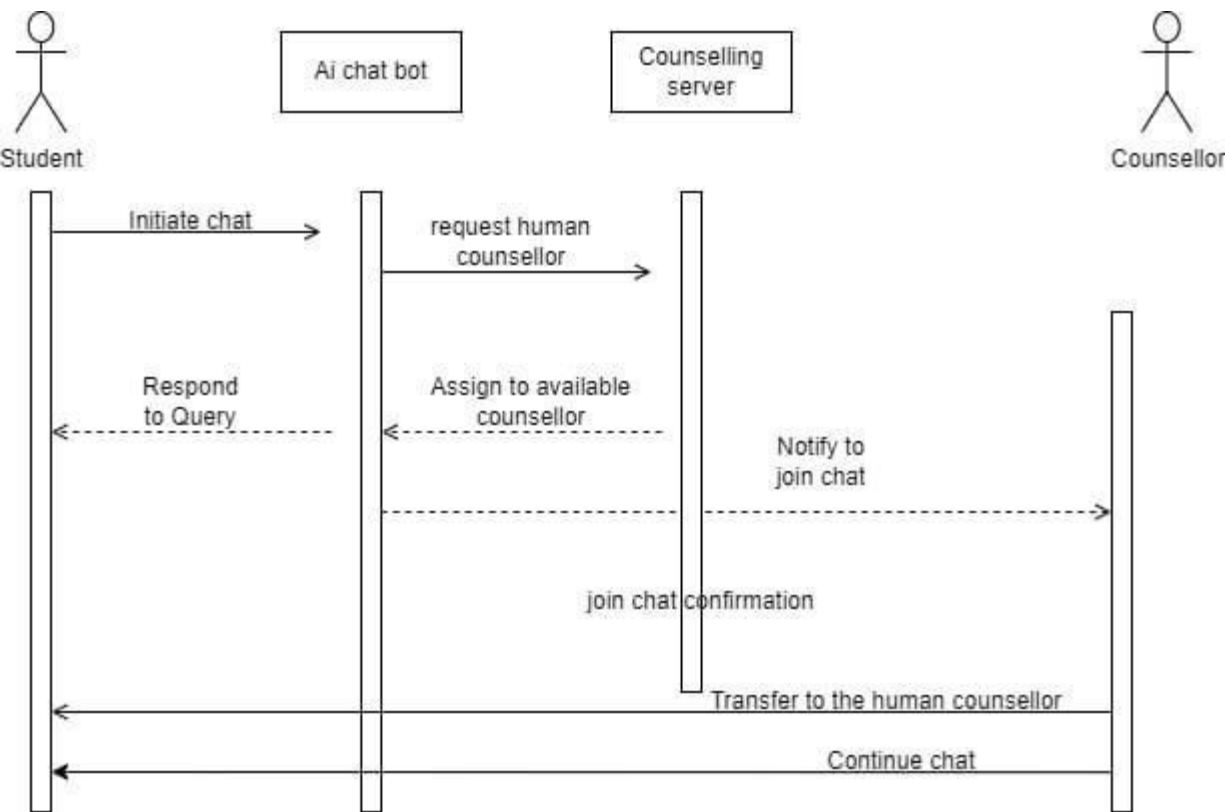
- A student enrollment to a project



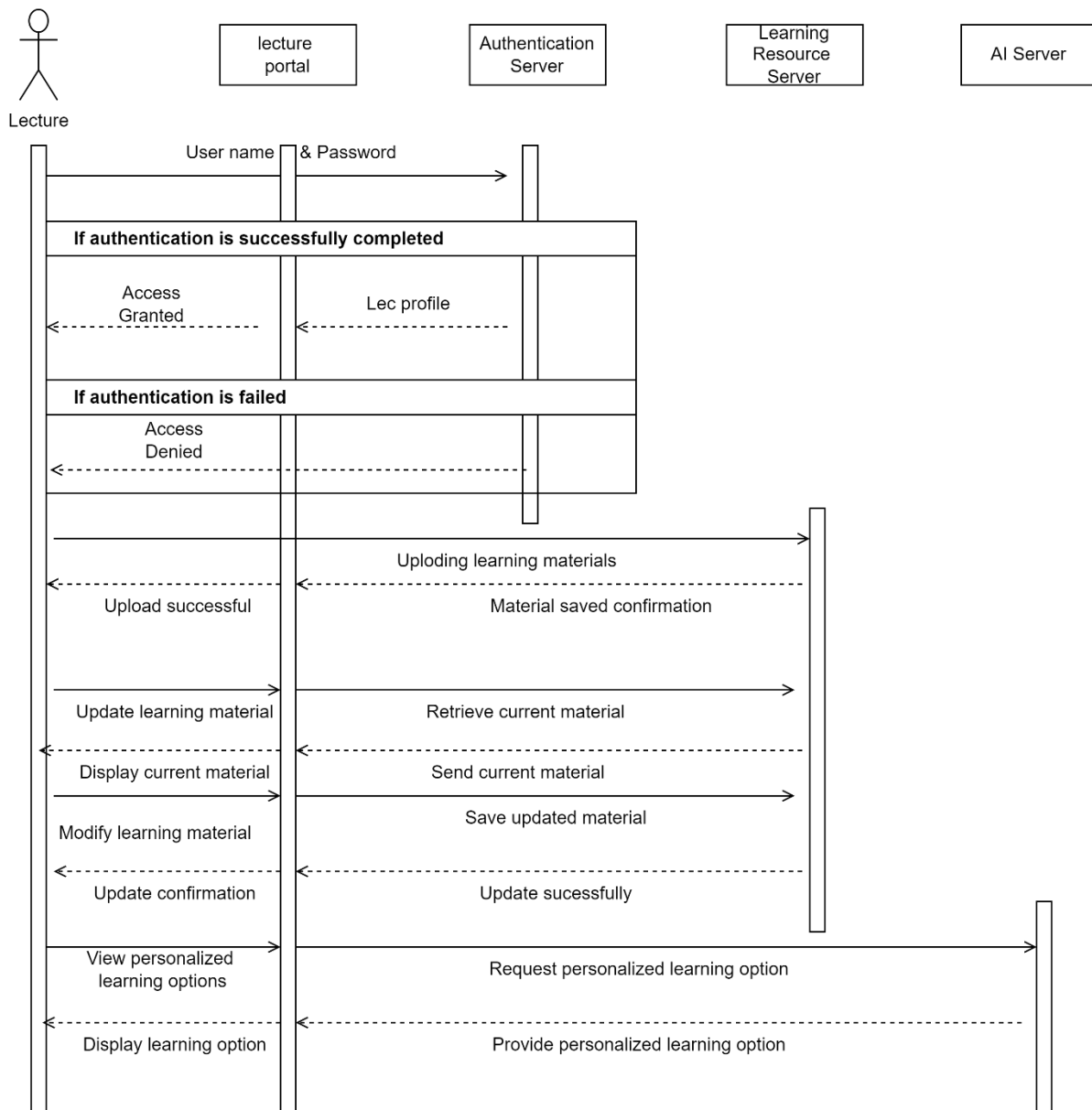
- Display of personalized learning modules

Sequence diagrams

Sequence diagram for AI chatbot

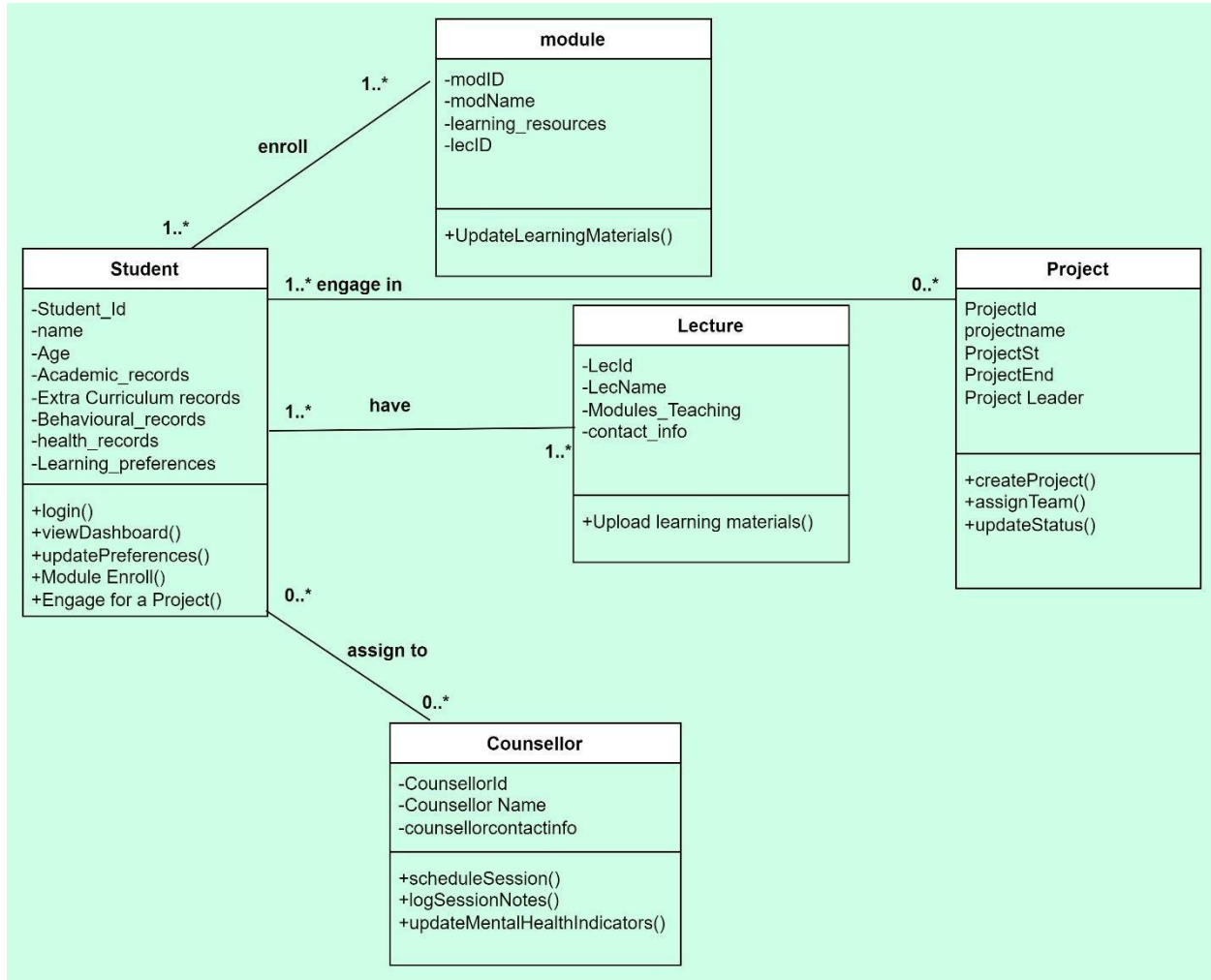


## Sequence diagram for Lectures actions



## Class Diagram

Class diagram for student engagement



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